



THE MAIN TITLE OF THE PROJECT

PROJECT ID

YOURNAME1,YOURNAME2,YOURNAME3,YOURNAME4

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WORD COUNT

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Churn Prediction in Dodge the Mud Game

PROJECT ID

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1. Problem definition

Dodge the Mud is a free mobile game in which players attempt to avoid falling mud. This project focuses on predicting player churn using gameplay data collected from January 2015 to May 2016, consisting of 163,929 rows and three columns: device, score, and time. The time variable indicates the Unix Epoch timestamp for each gameplay session.

In the context of mobile games, churn refers to a player who permanently stops playing (Runge & Gao, 2014). To construct reliable labels for churn and non-churn, two-time windows are required: the Observation Period (OP) and the Churn Prediction Period (CP). The OP begins at the player's first recorded session and continues for a defined number of days, during which the player's early behavior is observed. The CP follows the OP and serves as the period for determining churn status. A player with at least one session within the CP is considered a non-churner, while the absence of activity classifies the player as a churner. These definitions form the foundation for building ground-truth labels.

2. Methods

This section outlines the methods that are used in this project.

2.1 Preprocessing Data

Before building predictive models, it was necessary to convert timestamps into standard datetime format and examine the dataset in detail. Only nine observations contained missing device IDs, which represented a negligible portion of the dataset and were therefore removed. Exploratory data analysis included investigations of player lifetimes, retention rates, score patterns, and session behavior. These insights informed the selection of OP and CP windows and guided the subsequent feature engineering process.

2.2 Churn/no-churn label Creation

The choice of OP and CP lengths plays a critical role in defining meaningful churn labels. Long windows risk misclassifying genuine churners, whereas overly short windows may incorrectly label players who take short breaks. Based on behavioral analysis of gameplay patterns, several OP/CP configurations were tested to determine appropriate labeling schemes.

2.3 Feature Engineering

Feature engineering focused on capturing early player behavior during the OP. Both basic and advanced behavioral features were created, including session counts, score statistics, and measures of engagement. In total, 34 features were generated.

Feature selection proceeded through several stages. Low-variance filtering showed that all features exceeded the 1% variance threshold. Correlation analysis identified 13 feature pairs with correlations above 0.9; features with lower importance (based on Random Forest rankings) were removed from these pairs.

2.4 Models training

The dataset exhibited substantial class imbalance. For OP = 7 and CP = 14, churners accounted for 93% of the labeled data. Applying SMOTE to the entire dataset would produce excessive synthetic samples, potentially distorting the true data distribution. To mitigate this issue,

SMOTE was applied to only half of the data, yielding a more balanced distribution of 60% churn vs. 40% non-churn.

The data was then split into training and test sets, and models were evaluated using five-fold cross-validation. Three models were selected for training:

- **Random Forest:** A Random Forest is an ensemble of decision trees where each tree is trained on a bootstrapped sample of the data, and predictions are made through majority voting across trees (Breiman, 2001)
- **XGBoost:** a gradient boosting framework that constructs trees sequentially, with each tree trained to reduce the residual errors left by the previous ensemble (Chen & Guestrin, 2016).
- **CatBoost:** a gradient boosting algorithm designed to handle categorical features efficiently and reduce overfitting through ordered boosting and target statistics, leading to strong performance with minimal preprocessing (Dorogush, Ershov & Gulin, 2018).

These models were chosen due to their strong performance on tabular data and their ability to capture non-linear relationships.

2.5 Hyperparameter Tuning

Each model underwent hyperparameter optimization using randomized search approaches. The tuning process prioritized ROC-AUC and PR-AUC as the primary metrics due to the imbalanced nature of the dataset.

2.6 Evaluation and Error Analysis

Model evaluation considered accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC.

3. Results

3.1 Test OP/CP Combination

To determine a suitable combination of OP and CP, several configurations ranging from (OP = 1, CP = 3) to (OP = 7, CP = 14) were evaluated using stratified five-fold cross-validation with cost-sensitive learning.

The configuration (OP = 7, CP = 7) achieved the best results, with a ROC-AUC of 0.792 and a PR-AUC of 0.979, while maintaining a dataset of 25,956 samples and capturing 95% of true churners. Although this configuration produced the strongest metrics, it also resulted in an extremely imbalanced dataset, where 95% of players were labeled as churners. Such imbalance can inflate performance artificially by allowing models to predict the majority class. Therefore, the second-best configuration, OP = 7 and CP = 14, was selected because it offered strong performance while maintaining a more realistic class distribution.

3.2 Model Comparison (after tuning)

Overall, the evaluation of three ensemble learning models, Random Forest, XGBoost, and CatBoost, demonstrates consistently strong performance across all key metrics, with only subtle differences distinguishing them. The comparison is visualized in Figure 1. Error analysis showed that most misclassifications occurred among borderline cases, particularly players with minimal or inconsistent activity during the CP.

Random Forest achieved an accuracy of 0.941, precision of 0.950, and recall of 0.962, resulting in an F1-score of 0.956. Its ROC-AUC of 0.976 and PR-AUC of 0.985 highlight excellent discriminative ability and precision-recall balance, confirming that the model is highly effective at correctly identifying positive cases while minimizing false positives.

XGBoost slightly outperformed Random Forest in overall accuracy (0.951) and recall (0.971), leading to a higher F1-score of 0.964. Its ROC-AUC of 0.976 and PR-AUC of 0.981 indicate robust classification power, though its PR-AUC is marginally lower than Random Forest's. This suggests XGBoost is particularly strong at capturing true positives, making it well-suited for

tasks where recall is critical.

CatBoost delivered the highest accuracy (0.953) and precision (0.962), with a recall of 0.968 and an F1-score of 0.965. Its ROC-AUC (0.975) and PR-AUC (0.981) are nearly identical to XGBoost, reflecting balanced performance. Overall, all three models demonstrate excellent reliability, with CatBoost showing a slight edge in accuracy and precision, while XGBoost excels in recall.

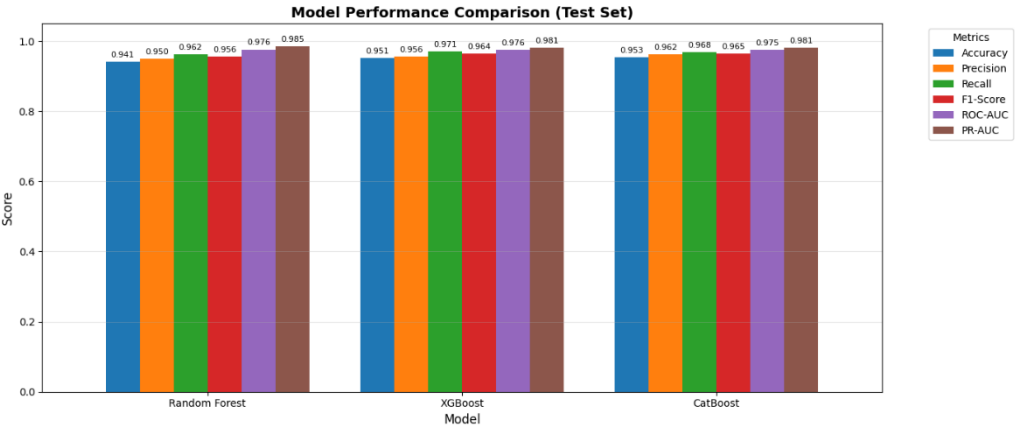


Figure 1: Model Comparison

3.3 Best Parameters

The best parameter of the best-performing model is shown in the table below.

Model	Parameter
Catboost	learning_rate : 0.1
	l2_leaf_reg : 1
	iterations : 400
	depth : 11
	border_count : 64
	bagging_temperature : 0

3.4 SHAP Analysis

Catboost is selected as the best-performing model as a result of the above findings, and SHAP analysis is performed on this model’s outcome. Figure 2 shows the top 10 features distributional impact, while Figure 3 demonstrates their importance values and rankings.

The features *weekend_ratio*, *night_play_ratio*, and *active_days_ratio* emerge as the most influential, exhibiting large mean absolute SHAP values, wide horizontal dispersions, and clear color gradients. This combination indicates not only strong overall impact but also substantial variability in how these features affect different users. Temporal behavior indicators, such as *avg_hour_of_day* and *peak_hour*, display consistent directional patterns, with higher values tending to push predictions upward and lower values pulling them downward, suggesting that later-day activity is particularly predictive of the target outcome. Features related to recency and behavioral diversity, including *recency_days* and *hour_entropy*, show narrower spreads yet still provide meaningful and stable contributions, pointing to consistent patterns in user behavior. Finally, engagement-volume and consistency features, such as *total_sessions*, *activity_days*, and *first_vs_avg_ratio*, exert moderate but reliable influence, reflected in their more compact SHAP distributions. Together, these patterns highlight the multifaceted behavioral signals driving the model’s predictive decisions.

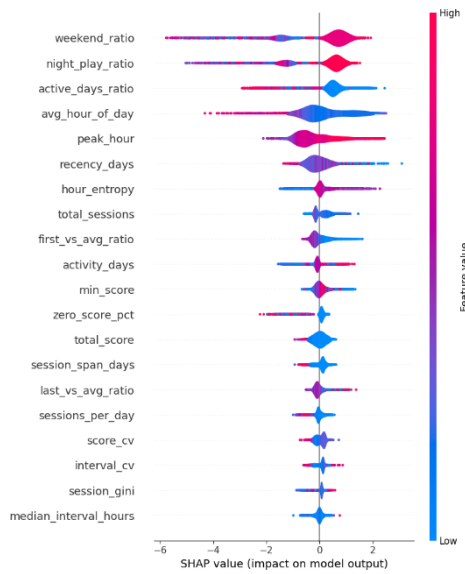


Figure 2: SHAP Beeswarm

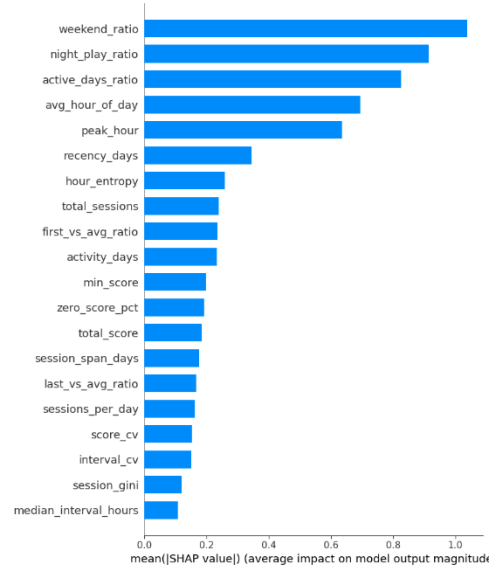


Figure 3: SHAP Importance Ranking

4. Discussion

The results demonstrate that early player behavior in *Dodge the Mud* provides a strong basis for predicting long-term churn. CatBoost emerged as the best-performing model, which aligns with its strength in handling complex interactions and categorical features. However, Random Forest and XGBoost also delivered high performance, indicating that the problem is well suited to tree-based ensemble methods.

The main methodological challenge involved addressing the extreme class imbalance inherent in many OP/CP combinations. The choice of OP = 7 and CP = 14 offered an effective balance between performance and label reliability. This experiment highlights the importance of thoughtful window selection in churn prediction tasks, as labeling strategies directly influence both model behavior and interpretability.

Feature engineering also played a critical role. While a large number of features were initially created, only a small subset contributed meaningfully to model performance, demonstrating that player behavior during the early stages of the game is relatively simple and can be captured effectively by a compact set of features.

5. Conclusion

This study successfully developed a predictive model for identifying player churn in *Dodge the Mud*. After evaluating several OP/CP configurations, mitigating class imbalance, engineering relevant features, and testing multiple tree-based models, CatBoost demonstrated the best overall performance. The findings show that player activity within the first seven days is highly informative for predicting future churn and that careful design of labeling windows is crucial for building robust churn prediction models. These insights can support game designers in developing more effective retention strategies and tailoring interventions for at-risk players.

6. References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

Chen, T., & Guestrin, C. (2016). Xgboost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.

Dorogush, A. V., Ershov, V., & Gulin, A. (2018). *CatBoost: gradient boosting with categorical features support*. *Advances in Neural Information Processing Systems (NeurIPS) Workshop*.

Declarations statement

State here what each team member contributed toward completing the project, indicating the name and the surname of the team member where applicable.

Code (Preprocess, EDA, Train models): Le Anh Thu Pham

Code (Hyperparameter tuning, OP/CP comparison, Visualization, Feature importance): Lam Minh Triet Thien

Writing (Problem statements, Methodology, Conclusion): Le Anh Thu Pham

Writing (Results, Discussion): Lam Minh Triet Thien

State here your use of Generative AI tools (which specific ones were used, if any, and what you used them for).

- Use Claude to make some visualizations better, to clean the code, fix bug.
- Use ChatGPT to suggest methods to handle the class imbalance, and we tried different methods and went with SMOTE for half dataset